

RAMCO INSTITUTE OF TECHNOLOGY
(Approved by AICTE, Affiliated by Anna University Chennai)
ACADEMIC YEAR 2025 – 2026
DEPARTMENT OF INFORMATION TECHNOLOGY REGULATION-2021
Course Outcomes

Semester I	
Course Code and Name of the course	Outcomes (CO) After completion of the course, the students will be able to
HS3151 Professional English I	CO1: Communicate clearly both in the written and oral form using appropriate vocabulary and comprehend written texts to make inferences. CO2: Speak persuasively in academic/work contexts and write biographical details and technical documents cohesively, coherently and flawlessly using appropriate words. CO3: Read, write and speak effectively for a variety of professional and social settings. CO4: Comprehend descriptive, narrative, expository and interpretive texts and write using creative, critical, analytical and evaluative methods. CO5: Understand and respond to different spoken and written discourses/excerpts in different accents and write different genres of texts adopting various writing strategies. CO6: Detect paralinguistic cues such as postures, gestures, facial expressions, and eye contact.
MA3151 Matrices and Calculus	CO1: Use the matrix algebra methods for solving practical problems. CO2: Apply differential calculus tools in solving various application problems. CO3: Use differential calculus ideas on several variable functions. CO4: Apply different methods of integration in solving practical problems. CO5: Apply multiple integral ideas in solving areas, volumes and other practical problems. CO6: Use MATLAB for finding eigenvalues of a Matrix, derivative and integration of functions.
PH3151 Engineering Physics	CO1: Understand the importance of mechanics. CO2: Express their knowledge in electromagnetic waves. CO3: Demonstrate a strong foundational knowledge in oscillations, optics and lasers. CO4: Understand the importance of quantum physics.

	CO5: Comprehend and apply quantum mechanical principles towards the formation of energy bands. CO6: Build a simulation models to calculate the energy values of particle in a 1D, 2D & 3D boxes using computer programming.
CY3151 Engineering Chemistry	CO1: Illustrate the importance of water quality parameters, water treatment methods and boiler troubles for domestic and industrial use. CO2: Explain the types of nanomaterials, synthesis and its applications in modern day devices. CO3: Apply the use of phase rule in metallurgy and describe the basics of composites and its industrial applications. CO4: Articulate the fuel types, synthesis and its combustion characteristics. CO5: Portray the working principle of alternate energy resources, storage devices and their application in electric vehicles and the impacts of carbon footprint. CO6: Analyse the water quality parameters of water samples collected in and around the native area
GE3151 Problem Solving and Python Programming	CO1: Develop algorithmic solutions to simple computational problems. CO2: Develop and execute simple Python programs. CO3: Apply control structures, functions and string to write simple programs for solving problems. CO4: Use Python data structures - lists, tuples, dictionaries to represent complex data. CO5: Illustrate read and write data from/to files in Python programs.
GE3171 Problem Solving and Python Programming Laboratory	CO1: Develop diagrammatic solutions to simple computational problems. CO2: Develop and execute simple Python programs. CO3: Implement simple python programs using control structures, functions and strings. CO4: Use python lists, tuples, sets and dictionaries to represent complex data for solving real life problems. CO5: Utilize Python packages, file operations in developing software applications.
BS3171 Physics and Chemistry Laboratory	CO1: Understand the functioning of various physics laboratory equipment. CO2: Use graphical models to analyze laboratory data.

	CO3: Use mathematical models as a medium for quantitative reasoning and describing physical reality. CO4: Access, process and analyze scientific information. CO5: Solve problems individually and collaboratively. CO6: Analyze the quality of the water samples like hardness, alkalinity, chloride and dissolved oxygen content. CO7: Quantitatively examine the impurities present in solutions by electro-analytical techniques such as pH, conductometry. CO8: Estimate the iron content in the solution by potentiometry. CO9: Practice the synthesis of nanoparticles.
Semester II	
HS3252 - Professional English -II	CO1: Compare and contrast products and ideas in technical texts and write relevant texts. CO2: Identify and report cause and effects in events, industrial processes through technical texts. CO3: Analyse problems in order to arrive at feasible solutions and communicate them in the written format. CO4: Present their ideas and opinions in a planned and logical manner. CO5: Draft effective resumes in the context of job search. CO6: Detect paralinguistic cues such as postures, gestures, facial expressions, and eye contact.
MA3251 – Statistics and Numerical Methods	CO1: This course aims at providing the necessary basic concepts of a few statistical and numerical methods and give procedures for solving numerically different kinds of problems occurring in engineering and technology. CO2: Acquaint the knowledge of testing of hypothesis for small and large samples which plays an important role in real life problems. CO3: Introduce the basic concepts of solving algebraic and transcendental equations. CO4: Introduce the numerical techniques of interpolation in various intervals and numerical techniques of differentiation and integration which plays an important role in engineering and technology disciplines. CO5: Acquaint the knowledge of various techniques and methods of solving ordinary differential equations.

PH3256 – Physics for Information Science	CO1: Extend the knowledge on classical and quantum electron theories, and energy band structures. CO2: Acquire knowledge on basics of semiconductor physics and its applications in various devices. CO3: Express the knowledge on magnetic properties of materials and their applications in data storage. CO4: Associate the necessary understanding on the functioning of optical materials for optoelectronics. CO5: Elaborate the basics of quantum structures and their applications and basics of quantum computing.
BE3251 – Basic Electrical and Electronics Engineering	CO1:Determine various electrical parameters of basic electric circuits CO2: Illustrate the construction, working principle and applications of various electrical machines. CO3: Interpret the characteristics of various analog electronic devices. CO4: Explain the basic concepts of digital electronics. CO5: Describe the fundamental elements and working of various measuring instruments.
GE3251 – Engineering Graphics	CO1: Construct the conic sectional curves, cycloidal and involute curves. CO2: Solve practical problems involving projection of lines. CO3: Construct the projection of simple solids and free hand sketches of multiple views from pictorial views of objects. CO4: Sketch the Section of Solids and Development of lateral Surfaces of solids. CO5: Illustrate the isometric and perspective projections of simple solids. CO6: Sketch the real components of machine / building / circuit drawing based on the basic concept learned from the topics.
CS3251- Programming in C	CO1: Develop simple applications in C using basic constructs and control structures CO2: Demonstrate C programs using arrays and strings CO3: Develop and implement modular applications using functions and pointers CO4: Develop programs using structures and union CO5: Design applications using sequential and random-access file processing.

CS3271 - Programming in C Laboratory	CO1: Develop programs in C using basic constructs and apply control structures for solving problems. CO2: Implement programs in C using Arrays. CO3: Develop applications using strings, functions and pointers CO4: Develop applications using structures and union CO5: Develop applications in C using file processing.
Semester III	
MA3354 Discrete Mathematics	CO1: Apply the concepts needed to test the logic of a program. CO2: Apply the knowledge of the counting principles to solve problems in Computer Science and Data Science Engineering. CO3: Apply the knowledge of Graphs and graph terminology in the Engineering subjects like Computer Networks, Data Structures etc. CO4: Apply the concepts and properties of algebraic structures in the Engineering subjects like Network Security, Cryptography etc. CO5: Apply the concepts and significance of lattices and Boolean algebra in Computer science and Data science engineering. CO6: Examine the Isomorphism between the Graphs using MATLAB
CS3351- Digital Principles & Computer Organization	CO1: Design various combinational circuits using logic gates. CO2: Design sequential circuits and analyze the design procedures. CO3: Illustrate the fundamentals of computer systems and analyze the execution of an instruction. CO4: Analyse different types of control design and identify hazards. CO5: Identify the characteristics of various memory systems and I/O communication.
CS3352 - Foundations of Data Science	CO1: Perform the data science process. CO2: Utilize the statistical techniques to describe the data in various formats. CO3: Analyse the relationship between data using correlation and regression techniques. CO4: Use the python libraries for data wrangling. CO5: Apply visualization libraries in python to interpret and explore data.
CD3291- Data Structures and Algorithms	CO1: Explain the abstract data types.

	CO2:Design, Implement, and Analyse linear data structures such as lists, queues and stacks for different applications. CO3: Implement Sorting, Searching and hashing techniques for given problems CO4: Design, Implement, and analyze efficient tree structures to perform searching, indexing and sorting. CO5: Implement efficient graph algorithms to solve graph problems.
CS3391 – Object-Oriented Programming	CO1: Apply the concepts of classes and objects to solve simple problems CO2: Develop programs using inheritance, packages and interfaces CO3: Make use of exception handling mechanisms and multithreaded model to solve real world problems CO4: Build Java applications with I/O packages, string classes, Collections and generics concepts CO5: Integrate the concepts of event handling and JavaFX components and controls for developing GUI based applications.
CD3281 - Data Structures and Algorithms Laboratory	CO1: Design and implement ADTs for given application CO2: Implement Linear data structures for the different real-world applications CO3: Experimenting hash tables, searching and sorting algorithms CO4: Apply tree data structure for searching, indexing and sorting CO5: Use graph algorithm to solve graph problems
CS3381 - Object-Oriented Programming Laboratory	CO1: Design and develop java programs using object-oriented programming concepts. CO2: Develop simple applications using object-oriented concepts such as package, exceptions. CO3: Implement multithreading, and generics concepts. CO4: Create GUIs and event driven programming applications for real world problems. CO5: Implement and deploy web applications using Java.
Semester IV	
CS3452- Theory of Computation	CO1: Construct the Deterministic and Non deterministic finite automata for any given regular language

	CO2: Solve regular expression for any regular language and find equivalence of regular expression and finite automata. CO3: Construct pushdown automata for the context-free language and write context-free grammar. CO4: Construct a Turing machine for computational purposes. CO5: Identify the given problem is decidable or un-decidable
CS3491 – Artificial Intelligence and Machine Learning	CO1: Use appropriate search algorithms for problem solving CO2: Apply probabilistic reasoning under uncertainty CO3: Build supervised learning models CO4: Apply ensembling and unsupervised models for real world applications. CO5: Build deep learning neural network models
CS3492 – Database Management Systems	CO1: To enable the students to learn the fundamentals of data models, relational algebra and SQL. CO2: To emphasis the students to represent a database system using ER diagrams and to learn normalization techniques. CO3: To cultivate students to understand the fundamental concepts of transaction, concurrency and recovery processing. CO4: To familiarize the students with the internal storage structures using different file and indexing techniques which will help in physical DB design. CO5: To expose the students to have an introductory knowledge about the Distributed databases, NOSQL and database security.
IT3401 – Web Essentials	CO1: Gain knowledge on various networking devices, web protocols and web programming CO2: Design interactive web pages using HTML5 and CSS3 CO3: Build dynamic web page with validation using Java Script objects and by applying different event handling mechanisms. CO4: Design and deploy simple web applications using PHP CO5: Develop interactive web applications using Servlet.
CS3451 – Introduction to Operating Systems	CO1: Learn the basic concepts and functions of operating systems. CO2: Make the students familiar with Scheduling algorithms and Deadlocks. CO3: Explore the students to learn various memory management schemes.

	CO4: To understand I/O management and File systems. CO5: Familiar with the basics of Linux system and Mobile OS like iOS and Android.
GE3451 – Environmental Science and Sustainability	CO1: Illustrate the importance of environment, biodiversity and its impact on human life. CO2: Correlate the causes, effect and control measures of various types of pollution and its case studies. CO3: Educate the availability of renewable energy resources and scientific concepts/principles behind them. CO4: Articulate the concept of sustainability and management. CO5: Gain knowledge on sustainability practices and socio-economical change. CO6: Analyze the integrated themes of biodiversity, pollution control, waste management, energy resources, and sustainable approaches.
CS3461 – Operating System Laboratory	CO1: Enable the students to install windows operating system, learn UNIX commands, shell programming, and system calls. CO2: Make the students to implement CPU Scheduling algorithms, inter-process communication and mutual exclusion using semaphore. CO3: Facilitate the students to learn about implementing deadlock avoidance, deadlock detection algorithms and threading. CO4: Assist the students to implement paging, virtual memory using virtual memory simulator. CO5: Make the students familiarize on implementing various memory allocation methods and page replacement algorithms. CO6: To enable the students to implement various disk scheduling, File Organization and File Allocation Strategies. CO7: Equip the students to create Linux virtual machine using VMware.
CS3481 – Database Management Systems Laboratory	CO1: Enable the students to learn and implement important commands in SQL. CO2: Familiarize the students with usage of nested and joint queries. CO3: Cultivate the students to understand functions, procedures and procedural extensions of databases. CO4: Expose the students to design and implementation of typical database applications. CO5: Familiar with the use of a front-end tool for GUI based application development.

CS3491 - Artificial Intelligence and Machine Learning Laboratory	CO1: Use appropriate search algorithms for problem solving. CO2: Apply reasoning under uncertainty. CO3: Build supervised learning models. CO4: Build ensembling and unsupervised models. CO5: Build deep learning neural network models.
IT3401- Web Essentials Laboratory	CO1: Apply JavaScript, HTML and CSS effectively to create interactive and dynamic websites. CO2: Create simple PHP scripts. CO3: Design and deploy simple web-applications. CO4: Create simple database applications. CO5: Handle multimedia components.
Semester V	
CS3591 - Computer Networks	CO 1: Explain the basic layers and its functions in computer networks. CO 2: Implement the functions of various transport layer protocols. CO 3: Examine the functions of the various protocols in the network layer CO 4: Apply the routing algorithms for a given problem CO 5: Implement data link and physical layer functions for data transmission
IT3501 - Full Stack Web Development	CO1: Understand the various stacks available for web application development CO2: Use Node.js for application development CO3: Develop applications with MongoDB CO4: Use the features of Angular and Express CO5: Develop React applications
CS3551 - Distributed Computing	CO1: Explain the foundations of distributed systems CO2: Solve synchronization and state consistency problems CO3: Use resource sharing techniques in distributed systems CO4: Apply working model of consensus and reliability of distributed systems CO5: Explain the fundamentals of cloud computing
CS3691 - Embedded Systems and IoT	CO1: Understand the internal architecture of embedded processors, including core components such as registers, memory, and execution units. CO2: Apply knowledge to interface processors with I/O devices and manage real-time systems. CO3: Design embedded applications for real-world challenges. CO4: Demonstrate IoT Communication models, APIs, and protocols (Bluetooth, WiFi, ZigBee, GPS, GSM). CO5: Design IoT applications using Arduino/Raspberry Pi /open platform.

Professional Elective - I CCS334 -Big Data Analytics	CO1: Describe big data and use cases from selected business domains. CO2: Introduce NoSQL in big data management. CO3: Perform map-reduce analytics using Hadoop. CO4: Install, configure, and run Hadoop and HDFS. CO5: Use Hadoop-related tools such as HBase, Cassandra, Pig, and Hive for big data analytics
Professional Elective - II CCS370 – UI and UX design	CO1: Build UI for user Applications CO2: Evaluate UX design of any product or application CO3: Demonstrate UX Skills in product development CO4: Implement Sketching principles CO5: Create Wireframe and Prototype
Mandatory Course – I MX3083 – Film Appreciation	CO1: Illustrate how to use the components of films CO2: Elaborate about the evolution of film language CO3: Summarize the various theories of films CO4: Describe the trends in film making from an Indian and World perspective CO5: Illustrate the past and present of Indian films
IT3511 - Full Stack Web Development Laboratory	CO1: Design full stack applications with clear understanding of user interface, business logic and data storage. CO2: Design and develop user interface screens CO3: Implement the functional requirements using appropriate tool CO4: Design and develop database based on the requirements CO5: Integrate all the necessary components of the application
Honors Course – I CCS343 & Digital and Mobile Forensics	CO 1: Understand the digital forensics process and its principles CO 2: Summarize the legal and ethical considerations of digital crime and investigations. CO 3: Explain the importance of forensic readiness in digital investigations. CO 4: Apply proper extraction techniques to obtain digital evidence from a specific iOS device. CO 5: Apply proper extraction techniques to obtain digital evidence from a specific Android device.
Honors Course – II CCS366 & Software Testing and Automation	CO1: Understand the basic concepts of software testing and the need for software testing CO2: Design Test planning and different activities involved in test planning CO3: Design effective test cases that can uncover critical defects in the application CO4: Carry out advanced types of testing CO5: Automate the software testing using Selenium and TestNG

Semester VI	
CCS356 & Object Oriented Software Engineering	CO 1: Compare various Software Development Lifecycle Models CO 2: Evaluate project management approaches as well as cost and schedule estimation strategies. CO 3: Perform formal analysis on specifications CO 4: Use UML diagrams for analysis and design. CO 5: Architect and design using architectural styles and design patterns, and test the system
IT3681 & Mobile Applications Development Laboratory	CO1: Design and build simple mobile applications supporting multiple platforms. CO2: Apply various programming techniques and patterns to build mobile applications. CO3: Build real-time mobile applications for society/environment. CO4: Build gaming and multimedia based mobile applications. CO5: Build AI based mobile applications for society/environment following ethical practices.
Professional Elective - III CCS335 & Cloud Computing	CO1: Understand the design challenges in the cloud CO2: Apply the concept of virtualization and its types CO3: Experiment with virtualization of hardware resources and Docker. CO4: Develop and deploy services on the cloud and set up a cloud environment. CO5: Explain security challenges in the cloud environment.
Professional Elective – IV CCS354 & Network Security	CO1: Classify the encryption techniques. CO2: Illustrate the key management technique and authentication. CO3: Evaluate the security techniques applied to network and transport layer CO4: Discuss the application layer security standards. CO5: Apply security practices for real time applications
Professional Elective – V CCW332 & Digital Marketing	CO1: Explore digital marketing and its role in online brand development. CO2: Apply search engine optimization in marketing for effective online promotion. CO3: Develop the email marketing and mobile marketing integrate with social media. CO4: Apply social media for brand engagement and influencer marketing CO5: Analyze digital marketing strategies using various online tools

Honors Course - III CCS336 & Cloud Services Management	CO1: Explore the fundamental principles of cloud service management. CO2: Develop the skills needed to manage cloud service strategies. CO3: Implement core concepts of cloud service management CO4: Analyze various cloud pricing models for effective cloud service management. CO5: Apply cloud governance frameworks to assess cloud services with cost-effectiveness.
Honors Course – IV CCS339 & Cryptocurrency and Blockchain Technologies	CO1: Elaborate emerging abstract models for Blockchain Technology. CO2: Identify major challenges and technical gaps existing between theory and practice in the crypto currency domain. CO3: Explain Different protocols and consensus algorithms in Blockchain. CO4: Apply Hyperledger Fabric and Ethereum platform to implement the Block chain Application. CO5: Build various block chain applications.
Mandatory Course – II MX3088 & State, Nation Building and Politics in India	CO1: Illustrate the role of State and Politics CO2: Describe the distribution of power through State organs and the Idea of India. CO3: Explore the evolution of political power and India’s constitutional development. CO4: Explain federalism, national integration and their role in nation-building. CO5: Illustrate the challenges and future of nation- building and political change in India.
Open Elective – I OCE351 & Environment and social Impact Assessment	CO1: Elaborate about the necessity to study the impacts of development on the environment. CO2: Describe the screening and scoping of developmental projects for environmental social impact assessments and explain different methodologies for environmental impact prediction and assessment. CO3: Plan environmental impact assessments, environmental management plans and evaluate environmental impact assessment reports. CO4: Plan environmental impact assessments, environmental management plans and evaluate environmental impact assessment reports. CO5: Solve real-time engineering problems based on prior experience gained by involving them in case studies
Semester VII	
GE3791 & Human Values and Ethics	CO1: Identify the importance of democratic, secular and scientific values in harmonious functioning of social life CO2: Practice democratic and scientific values in both their personal and professional life. CO3: Find rational solutions to social problems CO4: Behave in an ethical manner in society CO5: Practice critical thinking and the pursuit of truth.

GE3754 & Human Resource Management	CO1: Students would have gained knowledge on the various aspects of HRM CO2: Students will gain knowledge needed for success as a human resources professional CO3: Students will develop the skills needed for a successful HR manager. CO4:Students would be prepared to implement the concepts learned in the workplace CO5:Students would be aware of the emerging concepts in the field of HRM
Open Elective – II AI3021 & IT in Agricultural Systems	CO1:Discuss about the applications of IT in remote sensing applications in precision agriculture. CO2:Explain the greenhouse can be automated using environmental control systems. CO3:Apply IT principles and concepts for management of field operations. CO4:Apply different weather prediction models, their inputs and their applications. CO5:Apply Information Technology used for e-governance in agricultural systems. CO6:Apply Information technologies in agriculture to develop applications
Open Elective – III OFD352 & Traditional Indian Foods	CO1:Explain food system evolution and its cultural, religious, and social roles. CO2:Compare traditional and modern food processing for efficiency, nutrition, and shelf life. CO3:Identify regional food patterns and related IPR issues CO4:Analyze production, packaging, and marketing of traditional foods with SHGs and SMEs CO5:Evaluate traditional foods for nutrition, safety, cost, and environmental impact vs. fast foods.
Open Elective – IV OHS352 & Project Report Writing	CO1: Write effective project reports. CO2: Use statistical tools with confidence. CO3: Explain the purpose and intension of the proposed project coherently and with clarity. CO4: Create writing texts to suit achieve the intended purpose. CO5:Master the art of writing winning proposals and projects.
Honors Course – V CCS345 & Ethics and AI	CO1:Learn about morality and ethics in AI. CO2:Acquire the knowledge of real time application ethics, issues and its challenges CO3:Understand the ethical harms and ethical initiatives in AI. CO4:Learn about AI standards and Regulations like AI Agent, Safe Design of Autonomous and Semi-Autonomous Systems CO5:Understand the concepts of Roboethics and Morality with professional responsibilities CO6:Learn about the societal issues in AI with National and International Strategies on AI.

Honors Course – VI CCS346 & Exploratory Data Analysis	CO1: Understand the fundamentals of exploratory data analysis CO2: Implement the data visualization using Matplotlib. CO3: Perform univariate data exploration and analysis. CO4: Apply bivariate data exploration and analysis. CO5: Use Data exploration and visualization techniques for multivariate and time series data.
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